UNIVERSITY OF TORONTO

DATA SCIENCES INSTITUTE

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Dataset: Early Stage Diabetes Risk Prediction — UCI ML Repository (n=520)

Diabetes (Type 2) – Why Does it Matters

THE SCALE OF THE PROBLEM

Diabetes is one of Canada's fastest-growing chronic diseases.

Diabetes Facts

THE COST OF INACTION

~2M	1 in 3	200 000	~$1.5B	~$30B
Living with Diabetes	Will Develop Diabetes	New Cases Yearly	Treatment Cost in ON	National cost

RESEARCH QUESTION

What are the top symptoms based on genders and age groups that can predict and identify individuals with high risk of diabetes earlier as well as early-stage diabetes risk to enable preventive care?

Our Process

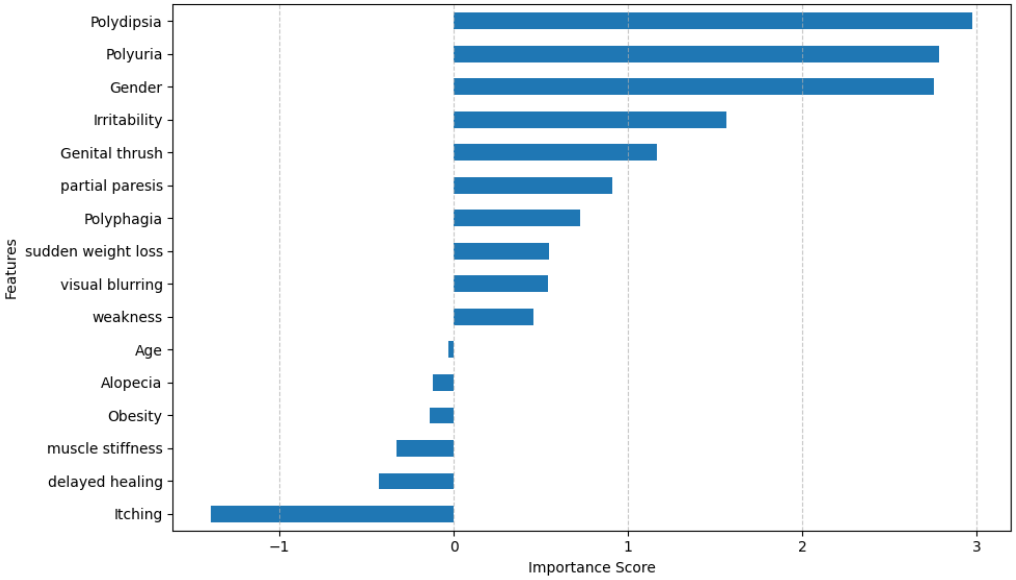
01 Data Preparation

02 Exploratory Data Analysis

03 Model-based Visualization

04 Communication

Top Predictors of Diabetes
(Feature Importance)



Diabetes Awareness Tool

IS YOUR BODY SENDING YOU EARLY WARNING SIGNS?

UNDERSTANDING SYMPTOMS THAT MAY BE LINKED TO DIABETES

Sometimes our bodies give subtle signals, like a "check engine" light.

Recognizing patterns early can help you take action sooner.

TOP 5 SYMPTOMS ASSOCIATED WITH DIABETES

85% Excessive Thirst (Polydipsia)

79% Frequent Urination (Polyuria)

Noticing These Symptoms? Talk To A Healthcare Professional.

51% Delayed Healing

48% Obesity

46% Visual Blurring

WHAT SHOULD YOU DO?

IF YOU NOTICE:

- Multiple symptoms
- Persistent changes
- Symptoms increasing over time

Consider discussing them with your healthcare provider.

Early screening is simple and can make a meaningful difference.

WHY THIS MATTERS

DIABETES OFTEN DEVELOPS GRADUALLY

Early symptoms may feel mild or easy to ignore.

But recognizing patterns early can help:

- Prevent complications
- Improve long-term health
- Start conversations with your doctor

Early detection can also reduce healthcare costs.

For Healthcare Professionals

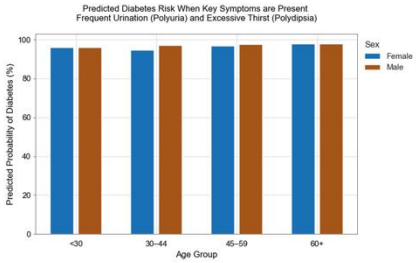
These findings highlight:

- Symptom clustering patterns
- Demographic variations
- Opportunities for structured screening tools

Visualization tools may support patient education and intake discussions

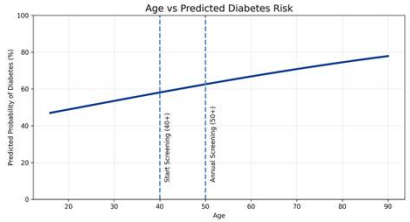
This flyer is for educational purposes only and is not a diagnostic tool. Please consult a healthcare professional for medical advice.

DIABETES RISK BY AGE, GENDER & SYMPTOMS



Frequent urination (polyuria) and excessive thirst (polydipsia) were common symptoms among both men and women diagnosed with diabetes. Some symptoms, such as muscle weakness and stiffness, showed slightly higher predicted risk among women.

DIABETES RISK BY AGE



Diabetes risk increases significantly after age 40. Regular screening can help detect it early and prevent complications.

Data Source: Early Stage Diabetes Risk Prediction Dataset (UCI Machine Learning Repository)

Thank You

for your attention

References:

1. York University: https://www.yorku.ca/science/mathstats/acadic/diabetes-in-toronto/
2. Frontiers in Public Health: https://www.frontiersin.org/journals/public-health/articles/10.3389/fpubh.2022.1029358/full)
3. ResearchGate: https://www.researchgate.net/publication/356828945_Use_of_Machine_Learning_Techniques_to_Predict_Diabetes_at_an_Early_Stage
4. UCI ML Repository: https://archive.ics.uci.edu/dataset/529/early+stage+diabetes+risk+prediction+dataset
5. Diabetes Canada (2019). New Data Shows Diabetes Rates And Economic Burden On Families Continue To Rise In Ontario: https://www.diabetes.ca/media-room/press-releases/new-data-shows-diabetes-rates-and-economic-burden-on-families-continue-to-rise-in-ontario--